

EXP:2 Support Vector Machine (SVM) and Random Forest for Binary & Multiclass Classification

Aim

To build classification models using **Support Vector Machine (SVM)** and **Random Forest**, apply them to the Iris dataset, and evaluate their performance using **accuracy** and **confusion matrix**.

Procedure

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Perform feature scaling using `StandardScaler`.
4. Split the dataset into training and testing sets.
5. Train the SVM classifier using `SVC()`.
6. Predict the test data using the SVM model.
7. Calculate the accuracy and display the confusion matrix.
8. Train the Random Forest classifier using `RandomForestClassifier()`.
9. Predict the test data using the Random Forest model.
10. Compare both models using confusion matrix heatmaps.

CODE:

```
!pip install pandas numpy matplotlib seaborn scikit-learn

import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data
y = iris.target

df = pd.DataFrame(X, columns=iris.feature_names)
df["Target"] = y

print("First 5 Rows of Dataset:")
print(df.head())

print("\nDataset Shape:", df.shape)
```

```

print("\nMissing Values:")
print(df.isnull().sum())

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

svm_model = SVC(kernel='linear')
svm_model.fit(X_train, y_train)

y_pred_svm = svm_model.predict(X_test)

print("\n==== SVM RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_svm))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

print("\n==== RANDOM FOREST RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))

plt.figure(figsize=(10, 4))

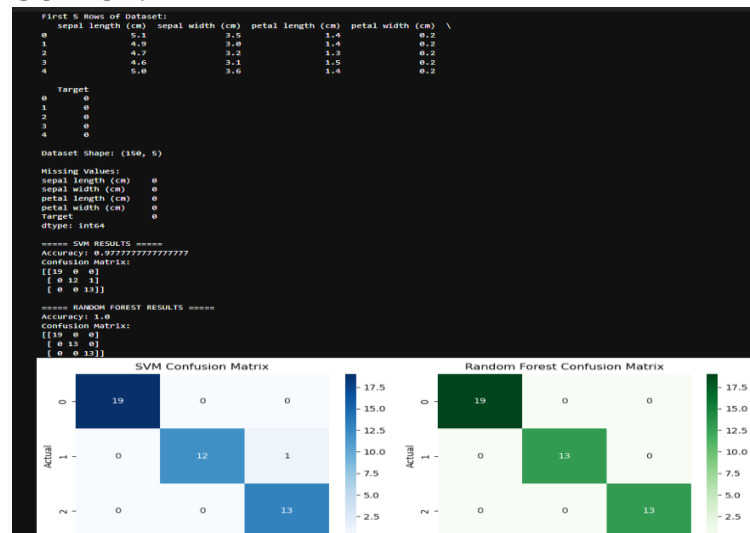
plt.subplot(1, 2, 1)
sns.heatmap(confusion_matrix(y_test, y_pred_svm),
            annot=True,
            fmt='d',
            cmap='Blues')
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

```

```
plt.subplot(1, 2, 2)
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True,
            fmt='d',
            cmap='Greens')
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.tight_layout()
plt.show()
```

OUTPUT:



Result

The **Support Vector Machine (SVM)** and **Random Forest** models were successfully implemented for binary and multiclass classification using the Iris dataset. Both models achieved high classification accuracy, and their performance was evaluated using confusion matrices and heatmaps.

